EMMANUEL OLUWAPELUMI OBOLO

Software Engineer

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Portfolio website Emmanuel Obolo Nigerian

ABOUT ME

Emmanuel is a highly driven Software Engineer currently studying at the African Leadership University. He is committed to the pursuit of excellence and views education as the crucial catalyst for long-term social and economic progress.

He specializes in building and developing impactful applications, leveraging AI/ML in the full SDLC to improve people's lives and contribute to global technology transformation.

WORK EXPERIENCE

09/2023 – 09/2025
Kigali, Rwanda

African Leadership University
Teaching Assistant - Mathematical Foundations of Machine Learning
- Guided students in applying linear algebra, calculus, and probability to optimize algorithms and train models while providing academic support through tutoring and problem-solving sessions to improve comprehension.

01/2025 – 01/2025
Kigali, Rwanda

Raise Her Voice Conference
Event Volunteer
- Managed registration and protocol logistics, ensuring a seamless check-in process while supporting event operations with strong organizational and communication skills.

05/2024 – 08/2024
Kigali, Rwanda

Africa Quantitative Sciences (AQS)
Full-Stack Engineer Intern
- Designed intuitive user interfaces, improving data visualization for over 50 users across departments.
- Leveraged front-end and back-end technologies to create 2 data-driven solutions that transform unstructured data into actionable

09/2023 – 12/2023
Nairobi, Kenya

Kadar Initiative for Community Empowerment
IT Assistant
- Maintained and updated the organization's blog daily for three months, enhancing online engagement while collaborating with a colleague to ensure accurate and timely content publication.

EDUCATION

09/2019 – 09/2022
Lagos, Nigeria.

BSc. Biochemistry (not completed)
University of Lagos
The Bachelor of Science in Biochemistry is designed to prepare students to have the basic general knowledge and skills in various areas of Biochemistry which they can develop to function as Biochemists in Medicine, Biotechnology, Industries, Agriculture, Environmental Science and Biological research areas.

01/2023 – 12/2025
Kigali, Rwanda

BSc. (Hons) Software Engineering
African Leadership University
Grade: Distinction

Thesis: The CBC English Proficiency Coach: A Grammar Correction Tool for Rwandan Education
Selected courses: Mathematics for Machine Learning, Introduction to Machine Learning, Machine Learning Techniques 1&2, etc.

SKILLS

Language skill ●●●●●
English

Language skill ●●○○○
Deutsch

Technical skills ●●●●○
Python, JavaScript.

Technical skills ●●●○○
Pandas, Matplotlib, Seaborn

Soft skills ●●●●○
Collaboration, Teamwork, Growth Mindset,
Continuous Learning, Problem Solving, Innovation.

Technical skills ●●●○○
C, Dart, SQL, Git, Linux

Technical skills ●●●○○
React, Flutter, Django, FastApi, Flask

Technical skills ●●●○○
Pytorch, Keras, TensorFlow, Scikit-Learn

Technical skills ●●●○○
PostgreSQL, SQlite, Firebase

INTERESTS

- Table Tennis
- Basketball
- Finance
- Reading
- Hiking
- Football

PROJECTS

11/2025 – 12/2025

Sparr AI
Survive The Interview
Sparr AI is a hyper-realistic mock interview platform that turns interview prep into a full-contact sport. Traditional interview prep tools are polite, scripted, and predictable. Real interviews are not. Candidates walk into high-stakes situations unprepared for: Pressure, Follow-up questions, and different personas.

Unlike passive interview simulators, Sparr AI combines two powerful technologies: ElevenLabs Conversational AI — Sub-second voice latency creates natural, flowing conversations Google Vertex AI (Gemini 2.5 Flash Lite) — Intelligent reasoning powers dynamic question generation.

01/2026 – 01/2026

KERN
kern was born from a real-world frustration during AI experimentation. While building a terminal-based conversational AI, the need to restart the entire session (and lose conversation context) every time a line of code changed became a bottleneck. This inspired the creation of a reloading system that could update logic on the fly without stopping the running script. Unlike web frameworks that restart the process, kern provides selective Script Reloading, allowing developers to modify logic and see results immediately in the same terminal session without losing state

08/2025 – 08/2025

V1 Template Engine
A personal project built to explore and understand the template engine by creating a simple lightweight python template engine.

01/2025 – 06/2025

V1 MVC Framework
A personal project built to explore and understand the Model-View-Controller architecture by creating a lightweight, modular framework for small-scale web applications.

Why This Framework?

- **Hands-on MVC Understanding:** Built to explore how small-scale MVC applications work.
- **Lightweight and Modular:** Each component is independent, making it easy to extend.
- **Practical Learning:** Focuses on applying theoretical MVC concepts in a structured manner.

11/2024 – 11/2024

Campus Cart

Campus Cart is a mobile application designed to simplify the process of finding and ordering food and products on campus. By providing a structured platform for vendors to list their offerings, students and faculty can easily browse, order, and connect with them, reducing the clutter associated with traditional WhatsApp group promotions.

The main goal of Campus Cart is to provide a streamlined platform for both buyers and vendors at ALU. By enhancing the way vendors and buyers communicate and transact, the app aims to eliminate the chaos of WhatsApp group promotions. This organization will not only increase sales for campus vendors but also offer greater convenience and accessibility for students and faculty, fostering a more vibrant campus marketplace.

09/2025 – 11/2025

The CBC English Proficiency Coach: A Grammar Correction Tool for Rwandan Education

The CBC English Proficiency Coach provides grammar correction with educational feedback aligned to Rwanda's Competency Based Curriculum, bridging a critical gap in grammar correction tools. Early models like FLAN-T5 and BART struggled with consistent, educationally relevant feedback, despite technical accuracy.

Mistral 7B Instruct emerged as the optimal solution, demonstrating superior instruction following and coherent educational explanation generation. This model achieved perfect BLEU and ROUGE scores, proving highly effective in delivering precise corrections and comprehensive, CBC-aligned feedback.

CERTIFICATES

AI Programming with Python (Udacity)
This Nanodegree covered the foundation of Python programming language, I also learnt about concepts needed as a foundation of building one's own neural network - Linear Algebra, Numpy, Pandas, Matplotlib, and PyTorch.

Calculus for Machine Learning and Data Science (DeepLearning.AI)
After completing this course, I was able to

• Analytically optimize different types of functions commonly used in machine learning using properties of derivatives and gradients
• Approximately optimize different types of functions commonly used in machine learning using first-order (gradient descent) and second-order (Newton's method) iterative methods
• Visually interpret differentiation of different types of functions commonly used in machine learning
• Perform gradient descent in neural networks with different activation and cost functions

Introduction to TensorFlow for Artificial Intelligence, Machine Learning, and Deep Learning (DeepLearning.AI)
This course teaches one how to use TensorFlow to implement deep learning principles so that one can start building and applying scalable models to real-world problems

AWS Machine Learning Fundamentals Nanodegree (Udacity)
This Nanodegree covers the fundamentals of Machine Learning and Deep Learning. I learnt to build intelligent systems using Python and popular data science libraries and gained hands-on experience by developing a machine learning workflow on a real-world dataset. By the end of the course, I had a solid understanding of the basics of Machine Learning and Deep Learning.

Neural Networks and Deep Learning (DeepLearning.AI)
In this course, I studied the foundational concept of neural networks and deep learning. By the end, I was familiar with the significant technological trends driving the rise of deep learning; build, train, and apply fully connected deep neural networks; implement efficient (vectorized) neural networks; identify key parameters in a neural network's architecture; and apply deep learning to my own applications.

Linear Algebra for Machine Learning and Data Science (DeepLearning.AI)
After completing this course, I was able to:

• Analytically optimize different types of functions commonly used in machine learning using properties of derivatives and gradients.
• Approximately optimize different types of functions commonly used in machine learning using first-order (gradient descent) and second-order (Newton's method) iterative methods.
• Visually interpret differentiation of different types of functions commonly used in machine learning.
• Perform gradient descent in neural networks with different activation and cost functions.

Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization (DeepLearning.AI)
In this course, the deep learning black box was opened to understand the processes that drive performance and generate good results systematically. By the end, I learnt the best practices to train and develop test sets and analyze bias/variance for building deep learning applications; was able to use standard neural network techniques such as initialization, L2 and dropout regularization, hyperparameter tuning, batch normalization, and gradient checking; implement and apply a variety of optimization algorithms, such as mini-batch gradient descent, Momentum, RMSprop and Adam, and check for their convergence; and implement a neural network in TensorFlow.

Publications

11/12/2025	**AI-Based Incentive Platform for Real-Time Dispatch of Flexibility Resources in Unlocking Grid Capacity**